

Towards modeling notions of pro-self vs pro-social attitudes in human-AI collaboration.

INTRODUCTION

- The Computers are Social Actors (CASA) paradigm (Nass et al., 1994) describes how humans implicitly apply the same social heuristics of human-human interactions to human-computer interactions.
- AI systems are ubiquitous, and racial biases in human-AI interactions reflect the lasting impact of racism on society.
- Expanded study by Atkins et al. (2021), who identified that racial biases influence cooperation with AI based on AI's perceived race by adding more subtly nuanced treatment conditions and building a cognitive model.
- Participants completed a human-AI cooperation task based on a pig chase game, as shown in Fig. 1, followed by a survey.
- Developed multiple ACT-R cognitive models (Anderson, 2007) using trust accumulation (Juvina et al., 2015) and instance-based learning (Gonzalez et al., 2003) to provide a cognitive-level, process-based explanation of the results.

RESEARCH QUESTIONS

- What strategies did participants use when they interacted with AI?
- What impact might the race of AI have on their strategy?
- Are these strategies transferable from the game to other real-world agents?

METHODS

- Over 950 participants.
- Rules:
 - Trust the AI to catch the pig for 25 points or exit for 5 points.
 - Played for 15 trials.
 - First 3 trials: Practice
 - 8th trial: Attention trial -> exit
- Participant demographics included:
 - Black/African American,
 - White/Caucasian
 - Non-White
- The AI used A* algorithm, not trained on human behaviors, same as Atkins et al.
- After completing 15 trials, participants answered five questions:

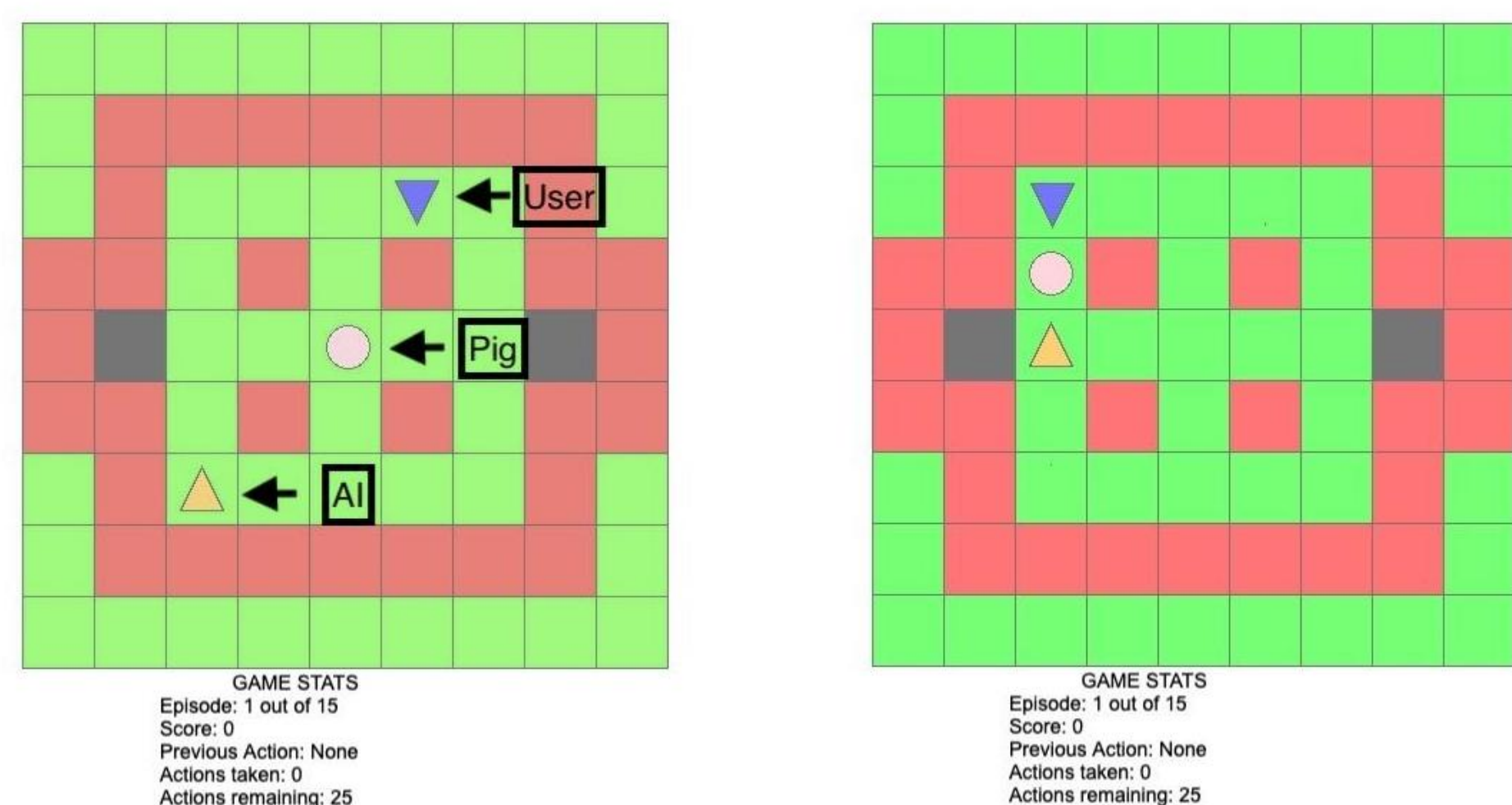


Fig. 1. Pig chase game with all pieces in the starting position on the left and the captured pig on the right.

RESULTS

Table 1. Participant demographics for each treatment condition.

Treatment	Group	Black	White	Non-White
B1	Black	47	48	48
B2	Black	46	47	44
BNP	Black	46	44	43
Control	Control	42	48	43
W1	White	44	48	44
W2	White	47	45	32
WNP	White	42	47	40

- After cleaning, 935 records were analyzed using a two-way ANOVA.
- Significant effects:
 - Participant Demographic $F(2,935) = 6.85, p < 0.005$,
 - Treatment X Demographic $F(2,935) = 2.22, p < 0.01$
- No significant effect for treatment alone $F(2,935) = 1.66, p = .12$ indicating an impact of demographic on the scores obtained.

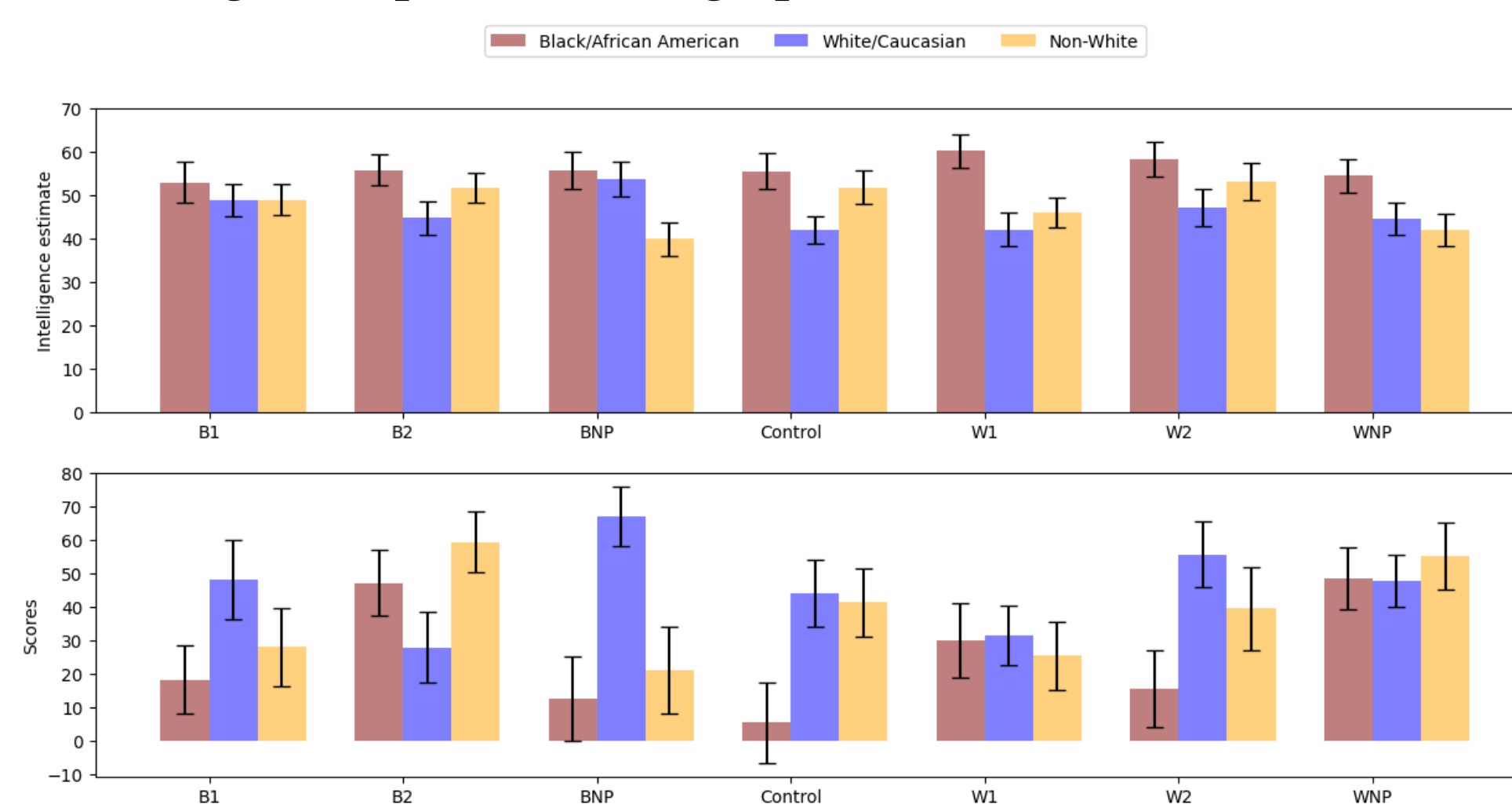


Fig. 2. Depicting the estimated intelligence of AI agents and average scores of the participants.

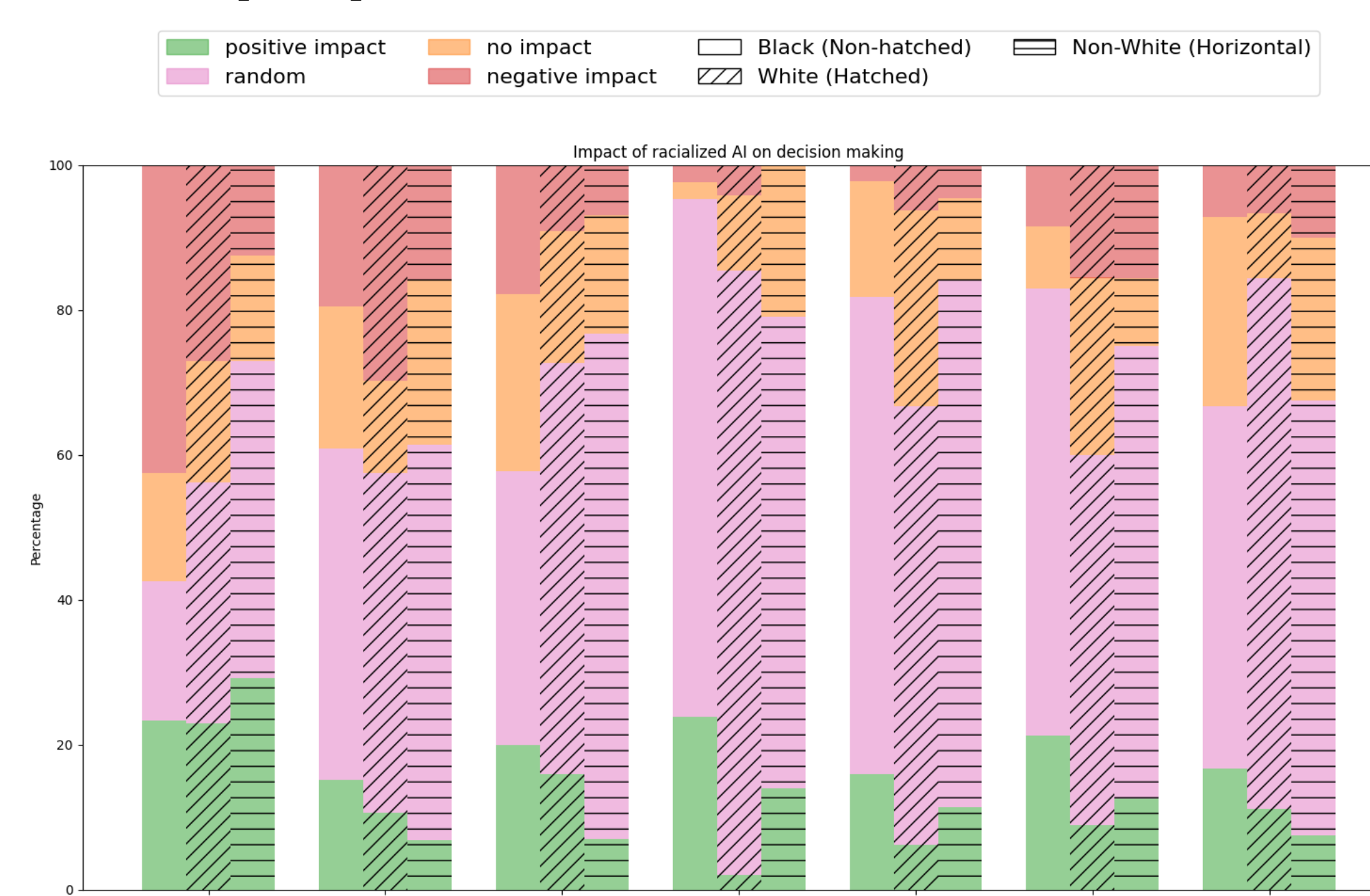


Fig. 3. Impact of racialized AI on decision-making

ACT-R MODEL

Model 1: The base model is a rational model from Dulam and Dancy (2025), where the human agent's cooperation with AI is based on a continued reduction in AI's proximity to the pig. This model is discussed in Figure 4.

Model 2: This model accumulates trust towards the opponent as the trials progress, similar to Juvina et al. (2015). The cumulative trust $T(i)$ is defined as follows:

$$T(i) = T(i-1) \pm [(N-n) * \alpha * \beta] - \gamma$$

where $T(0)$ is set based on the survey responses, $T(0) \in [0-4]$. N is total number of trials and n is current trial number, and α is a scaling parameter set to 0.01, β is guided by rule that triggers the update and γ is guided by *score_awareness*.

$$\beta = \begin{cases} 0.5, & \text{for decisions occurring at goal level} \\ 0.1, & \text{for decisions occurring at sub-goal level} \end{cases}$$

$$\gamma = AI_{exit} - score_{awareness}$$

Finally, if the AI agent is near one of two exits, the user might suspect it is exiting, which in turn reduces trust.

$$AI_{exit} = \begin{cases} 0.1, & \text{AI at exit} \\ 0, & \text{otherwise} \end{cases}$$

score_awareness is set to 0.1 whenever the score becomes negative, and actions remaining are less than 5. The model will *decide to exit* whenever the cumulative trust value drops below the threshold.

Model 3: Implements instance-based learning from Gonzalez et al. (2003) to retrieve a previous decision made in similar scenario to update the trust values.

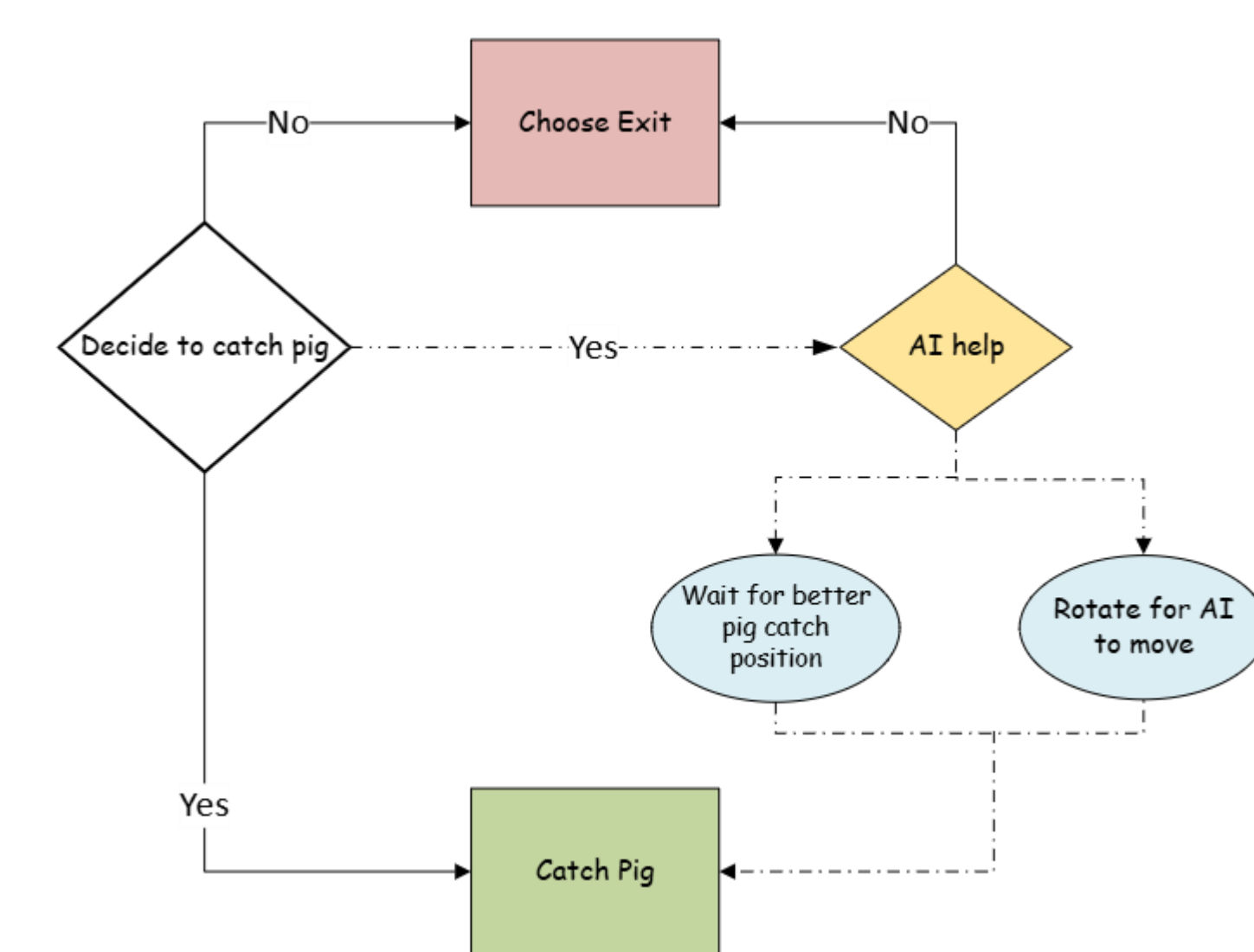


Fig. 4. Base model

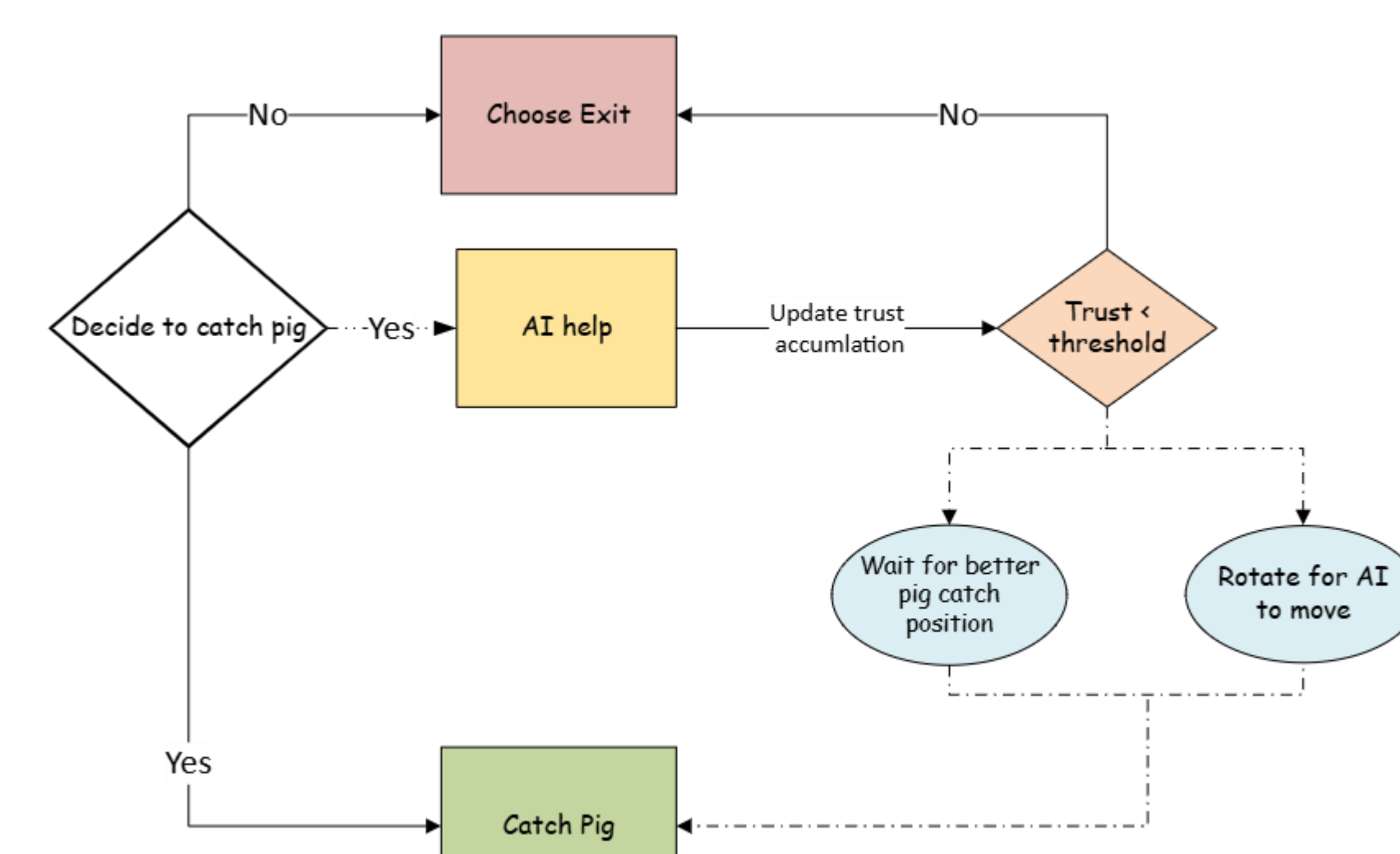


Fig. 5: Trust accumulator model

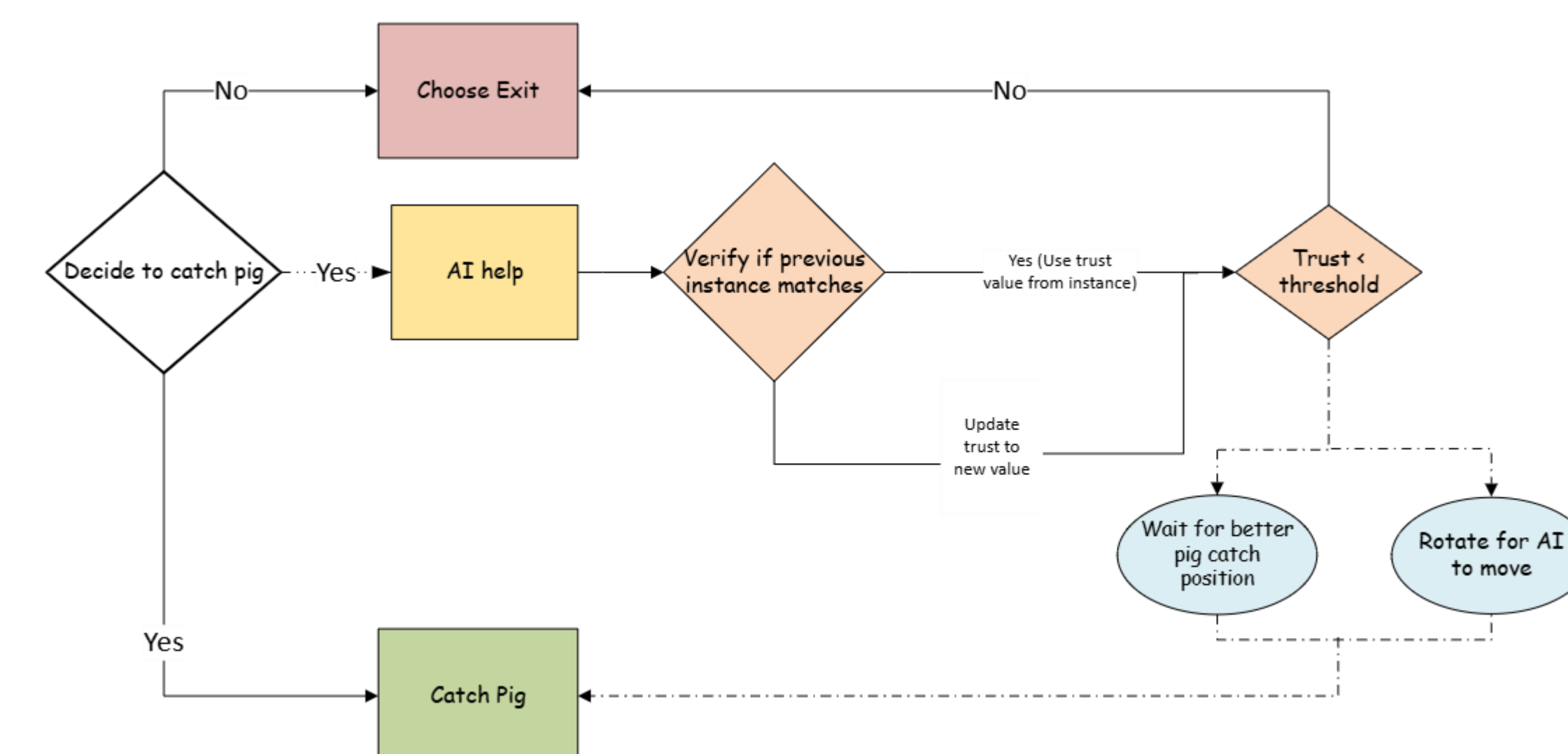


Fig. 6: Instance-based trust accumulator model

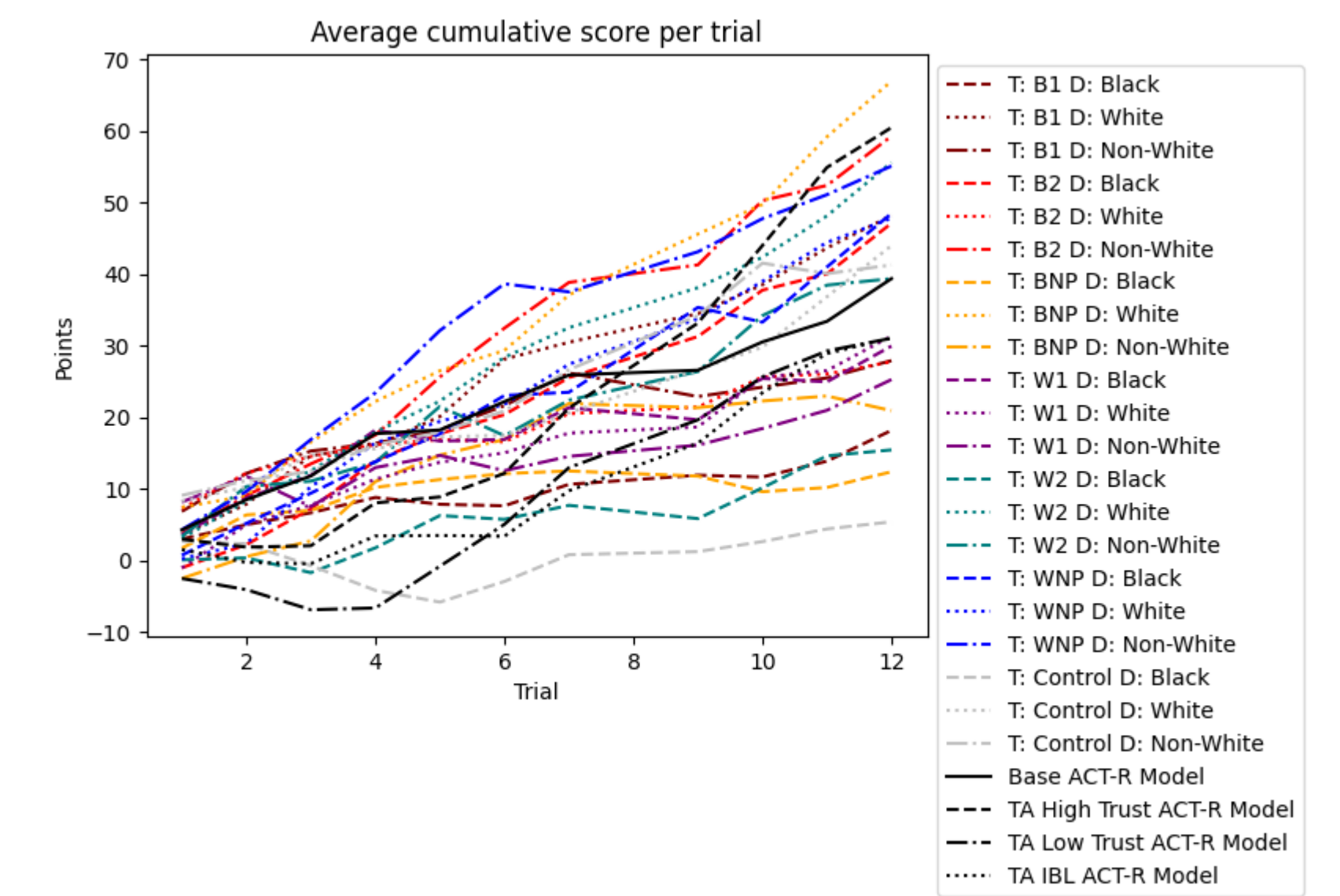


Fig. 7. Comparison of different ACT-R models with treatments.

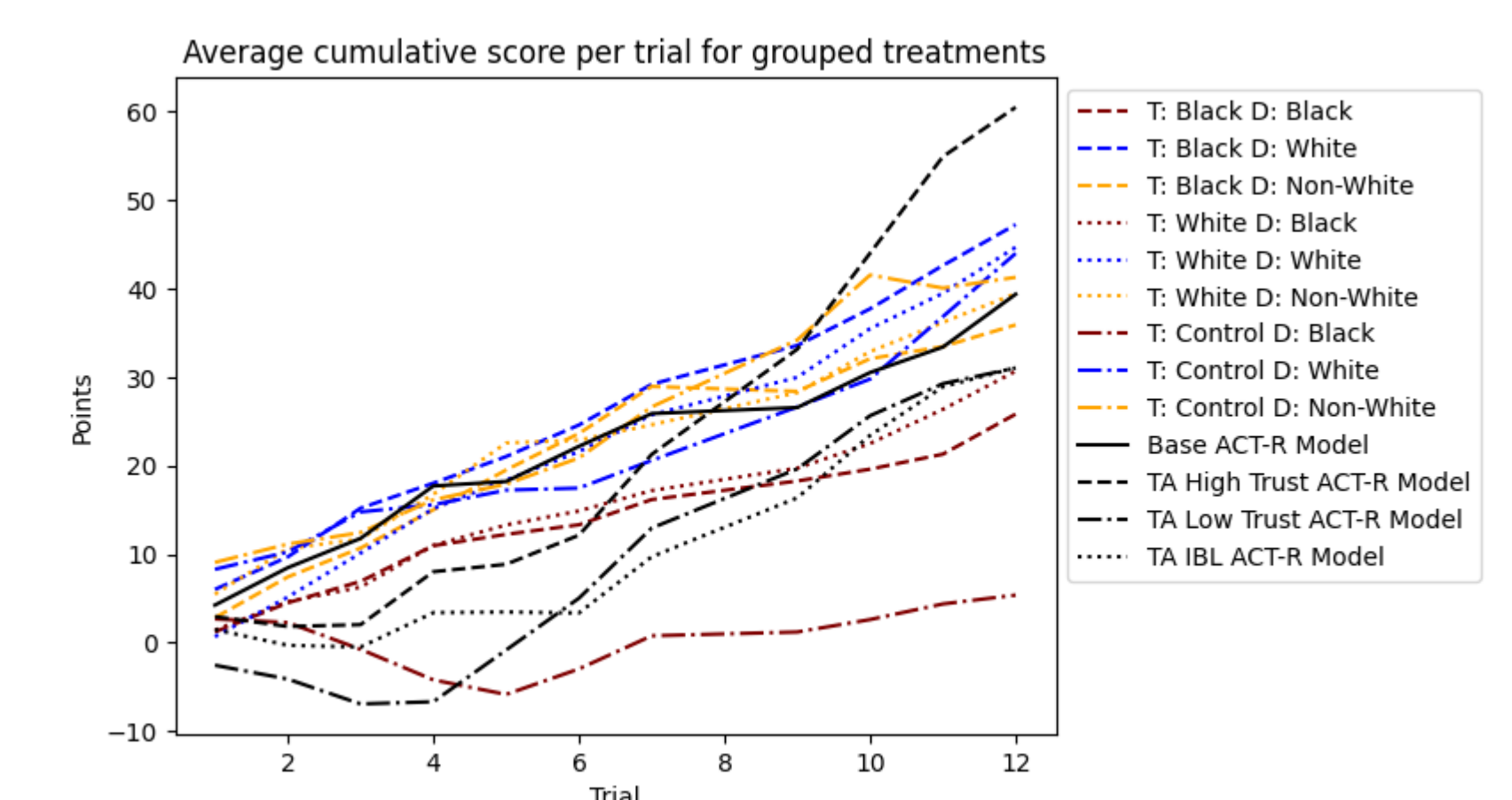


Fig. 8. Comparison of different ACT-R models with grouped treatments.

CONCLUSION & FUTURE WORK

- Race of AI agent impacted participant strategies, with people in the B1 treatment condition having the highest negative opinions about AI.
- The models achieve comparable fits to participant scores; performing a grid search in parameter space can help achieve individual treatment-level fit for 21 treatments.
- The model can be extended to include more individual-level modeling with Serdiuk & Ritter (2025).
- Implicit Association Task (IAT) could help explore individual implicit biases.

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